

Breakthrough Seizure Following Dose Escalation of Extended-Release Antiepileptic Neurotab XR: Case Report

J. Okafor MD (1), P. Singh MD (1) | (1) Dept. of Neurology, Test University Hospital

Abstract: We report a case of a 34-year-old man with pre-existing epilepsy who experienced a generalized tonic-clonic seizure shortly after dose escalation of extended-release Neurotab XR. The episode prompted hospitalization for observation; the patient stabilized after the dose was held. We discuss possible mechanisms and monitoring recommendations during dose titration. (Fictional case, created for software testing purposes.)

Background

Extended-release antiepileptic formulations are frequently titrated upward to achieve seizure control, and paradoxical breakthrough seizures during titration, while uncommon, have been described in the literature, particularly when dose increases are rapid or when adherence and absorption are inconsistent.

Case Report

A 34-year-old man with a longstanding history of epilepsy, previously well controlled on a stable regimen, was escalated from 150 mg to 300 mg twice daily of Neurotab XR as part of a planned titration schedule. Approximately two hours after his evening dose on day six of the new regimen, he experienced a generalized tonic-clonic seizure lasting approximately 90 seconds, with a post-ictal confusional period of about 20 minutes. He was transported to the emergency department and admitted for observation. The suspect medication was held during the admission and he was monitored with continuous EEG, which did not reveal ongoing epileptiform activity. He was discharged after 24 hours in stable condition with neurology follow-up arranged to reassess the titration plan.

Discussion

Breakthrough seizures during dose titration of antiepileptic medications may reflect transient pharmacokinetic fluctuations, particularly with extended-release formulations, rather than treatment failure per se. Careful monitoring during titration phases, along with patient education about seizure precautions, is recommended. Rechallenge decisions should be individualized.

Conclusion

This case highlights the importance of vigilant monitoring during antiepileptic dose titration and contributes a further data point to the sparse literature on breakthrough seizures temporally associated with dose escalation of extended-release formulations.

References

1. Perucca E. Pharmacokinetic variability of antiepileptic drugs during dose titration. *Epilepsia*. 2015;56(Suppl 1):S15-S22.
2. French JA. Refractory epilepsy: clinical overview. *Epilepsia*. 2007;48(Suppl 1):3-7.
3. Guidelines for Reporting Neurological Adverse Events in Clinical Practice, Test Neurology Society, 2021.